

RAMAAKSHAY MALLIREDDY

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OBJECTIVE

Highly motivated Aerospace Engineering student with experience in automation tools, embedded/ML programming, software development, hardware manufacturing, and 3D Modeling. I am eager to apply my technical background to real-world projects and contribute to the development of innovative, scalable engineering solutions.

EDUCATION

University of Texas at Austin

Bachelors in Aerospace Engineering

Courses:

Computer Programming (Python)

Engineering Computation with MATLAB

C++ Data Structures and Algorithms

Differential Equations & Linear Algebra

AI Design and Development

Machine Learning in Robotics

Austin, TX

Expected Graduation: May 2028

WORK EXPERIENCE

Austin Energy

Electrical Engineering Intern

Austin Energy HQ, Austin, TX

June 2024 - August 2024

- Utilized AutoCAD AUD to design 10+ on-site implementations, converting customer work orders to physical designs
- Sized and designed multiple electrical components, including three-phase transformers and protective relays
- Automated over 200+ electrical grid simulations maximizing run-time efficiency for existing automation frameworks

IBM

Platform/ML Engineering Intern

Austin, Texas

May 2026 – August 2026

- Designed AI/ML-powered hybrid cloud solutions for enterprise clients as part of IBM's multi-disciplinary Client Engineering team
- Utilized ECE foundations—systems architecture, computer networks, embedded systems, and hardware-software integration—to build reliable components within distributed cloud environments
- Implemented cloud-native workloads using Python/Java, containerization (Docker/Kubernetes), and orchestration on Red Hat OpenShift and IBM Cloud platforms.

RESEARCH EXPERIENCE

National High School Journal of Science

Scientific Research Conductor

Austin, TX

May 2025 – August 2025

- Conducted research, focusing on optimal methods of celestial body detection
- Designed experiments testing **10+** different methods of celestial body detection: telemetry, astrometry, radio wave detection, etc.
- Published research in a peer-reviewed journal, which has been cited and utilized as a data source in subsequent scientific experiments.

CLUB EXPERIENCE

Texas Spacecraft Laboratory (Electrical Power Systems Sub-team)

September 2025 – Current

- Selected and tested over **15+ power components**, including batteries, solar panels, and converters, ensuring compatibility with spacecraft power requirements.
- Developed and ran orbital energy simulations predicting power consumption within $\pm 5\%$ accuracy across mission scenarios.
- Designed and validated **3+ custom PCBs** supporting subsystem power distribution and DC-DC conversion, improving system efficiency by $\sim 10\%$.

AIAA Aircraft Design Team

Propulsion Engineer

August 2025 - Current

- Selected and integrated 2 high-thrust jet engines to achieve **>250 N** total thrust, enabling catapult-assisted launch and sustained Mach 1.6 dash performance for the DC25 aircraft.
- Modeled and optimized fuel and thermal management systems using MATLAB and ANSYS, reducing avionics operating temperature by 18% and increasing system endurance by 22%.
- Coordinated with aerodynamics and structures teams to refine engine nacelle geometry and intake placement, improving thrust-to-weight ratio by 15% and cutting drag by 9%.

PROJECT EXPERIENCE

Saturn V Replica Modeling

February 2025 – May 2025

- Created a detailed 3D model of the Saturn V rocket using Autodesk Inventor and AutoCAD
- Constructed an exact physical replica using custom plastic molds based on the CAD designs
- Applied principles of aerodynamics and physics throughout the design and applied FEAs & CFDs

SKILLS

Programming and Software: Python (PyTorch, Sentence Transformers, Pandas), C, C++, MATLAB, Java, GIT, Linux, SolidWorks, CAD, SQL, HTML

Other: Oscilloscope, waveform generator, soldering, Microsoft Office, academic research writing, presentation skills